PADHAKU
Curiosity · Class 6 Science



CHAPTER 7

Temperature and its Measurement

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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⭐ Introduction

? Can it always be correctly judged, only by touching a person, that the person has fever?

✔ ANSWER

No. Touching only gives a rough idea and our sense of touch can be wrong.

To be sure, we must measure the temperature with a **thermometer**.

⭐ 7.1 Hot or Cold?

Activity 7.1 **Dip your right hand in A (warm water) and left hand in C (ice-cold water) for 1–2 minutes, then put both hands in B (tap water). What will each hand feel? Write predictions, then observe.**

💡 SAMPLE ANSWER

- **Right hand** (from warm water A) feels the water in B **cool**.
- **Left hand** (from ice-cold water C) feels the same water in B **warm**.

Inference: The same water feels different to the two hands. So we cannot always rely on our sense of touch to decide correctly whether a body is hot or cold.

? **Then how do we find out how hot or cold a body is?**

✔ ANSWER

By measuring its **temperature** with a thermometer. Temperature is a reliable measure of hotness or coldness.

⭐ 7.3.1 Clinical thermometer

Do you know? **During the COVID-19 pandemic, special thermometers were used to measure temperature of a person from a distance. What were those?**

✔ ANSWER

They were **non-contact thermometers**, also called **infrared thermometers**.

They measure temperature without touching the body, so the risk of spreading disease is reduced.

Activity 7.2 **Measure your body temperature and that of friends with a digital clinical thermometer and record it in Table 7.1.**

💡 SAMPLE ANSWER

Steps: wash hands and thermometer tip with soap and water → press reset → place under the tongue, close mouth → wait for the beep/light → read the display → clean and dry the tip.

S. no.	Name	Temperature (°C)
1	Ravi	36.8
2	Meena	37.0
3	Arjun	37.2

Observation: Most readings are close to 37.0 °C, but some are slightly higher or lower.

? Did you find that the normal temperature of every person was 37.0 °C?

✔ ANSWER

No. Some people had a slightly higher or lower temperature.

37.0 °C is the **average** body temperature of a large number of healthy people. Body temperature depends on age, time of the day and activity level.

? Do small children generally have slightly higher body temperatures as compared to adults?

✔ ANSWER

Yes. Small children generally have a slightly higher body temperature than adults.

? **Do old people, even when healthy, generally have lower body temperatures than young adults?**

✔ ANSWER

Yes. Healthy old people generally have a slightly lower body temperature than young adults.

Try this **Measure your own temperature at different times of the day and on different days for a month. Are there any variations? What might be the reasons?**

💡 SAMPLE ANSWER

Yes, small variations are seen (for example 36.6 °C in the early morning and 37.2 °C in the evening).

Reasons: time of the day, physical activity (after playing or running), eating, weather.

? **A friend's digital thermometer shows the normal temperature of a healthy human body as 98.6 °F. What is the reason for this difference?**

✔ ANSWER

His thermometer uses the **Fahrenheit scale** (°F), not the Celsius scale (°C).

37.0 °C on the Celsius scale is the same as **98.6 °F** on the Fahrenheit scale.

? **How was fever detected before thermometers were developed?**

✔ ANSWER

By checking the **pulse rate**; fever affects the pulse rate. This was known even in olden days in India.

But other situations also change the pulse rate, so pulse rate alone is not a reliable indicator of fever.

? **Can a clinical thermometer be used for measuring the temperature of boiling water? Or of ice?**

✔ ANSWER

No. The temperatures of boiling water and ice are outside the range of a clinical thermometer (body temperature does not normally go below 35 °C or above 42 °C).

? **How can we measure temperatures beyond the range of a clinical thermometer?**

✔ ANSWER

By using a **laboratory thermometer**, which has a wider range (usually –10 °C to 110 °C).

⭐ 7.3.2 Laboratory thermometer

Activity 7.3 **Take a laboratory thermometer. Note the lowest and highest temperature it can measure. What is its range? Also tell the range of the thermometer in Fig. 7.3a.**

💡 **SAMPLE ANSWER**

- Lowest temperature: **–10 °C**
- Highest temperature: **110 °C**
- Range: **–10 °C to 110 °C** (same as Fig. 7.3a)

(Your school thermometer may have a different range; read its scale.)

Activity 7.4 **Find the smallest value a laboratory thermometer can read: temperature difference between two bigger marks, number of small divisions between them, value of one small division.**

💡 **SAMPLE ANSWER**

- Difference between two bigger marks (e.g. 0 °C and 10 °C): **10 °C**
- Number of small divisions between them: **10**
- One small division = 10 ÷ 10 = **1 °C**

So the smallest value the thermometer (Fig. 7.3b) can read is **1 °C**.

Activity 7.5 **Measure the temperature of warm water in a beaker with a laboratory thermometer. What happens when you take the thermometer out of the water?**

💡 **SAMPLE ANSWER**

Reading (example): about **45 °C**; friends' readings are nearly the same.

As soon as the thermometer is taken out, the liquid column starts to fall.

Conclusion: The temperature must be read while the thermometer is still in the water.

? **Can we use a laboratory thermometer for measuring the body temperature of a person?**

✔ **ANSWER**

No. The liquid column starts falling as soon as it is taken out of the mouth, so we cannot read it correctly. It is also long, breaks easily and is not meant to be put in the mouth.

More to do! **Observe the thermometers in the setups of ice and boiling water (made by your teacher). What are the**

temperatures of ice and boiling water? Do they change after some time?

💡 SAMPLE ANSWER

- Melting ice: about **0 °C**
- Boiling water: about **100 °C** (slightly less at high places like Shillong)

The readings do not change after some time: temperature stays constant while ice melts and while water boils.

Activity 7.6

Readings of boiling water in Shillong: Phiban 97.8 °C, Shemphang 98.0 °C, Onestar 97.9 °C, Kloi 98.0 °C, Bandarisha 98.1 °C. Why are there differences in their readings?

💡 SAMPLE ANSWER

Possible reasons (the correct way of reading was not followed):

- thermometer tilted, not held vertically
- bulb touching the bottom or sides of the beaker
- eye not in line with the top of the liquid column
- reading taken after taking the thermometer out, or too early

⭐ 7.3.3 Air temperature

Activity 7.7 **Record the maximum and minimum air temperature of a place for 10 days from weather reports (Table 7.3). Do they stay at the same level?**

💡 SAMPLE ANSWER

S.no.	Date	Maximum (°C)	Minimum (°C)
1	1 Nov	30	18
2	2 Nov	29	17
3	3 Nov	31	18

No. The maximum and minimum temperatures vary from day to day, because weather depends on several factors. They rise as summer comes and fall in winter.

🔭 Let us enhance our learning

1 **The normal temperature of a healthy human being is close to ____. (i) 98.6 °C (ii) 37.0 °C (iii) 32.0 °C (iv) 27.0 °C**

✔ ANSWER

(ii) 37.0 °C

It is the average body temperature of healthy people (98.6 is in °F, not °C).

2 **37 °C is the same temperature as ____. (i) 97.4 °F (ii) 97.6 °F (iii) 98.4 °F (iv) 98.6 °F**

✔ ANSWER

(iv) 98.6 °F

37.0 °C on the Celsius scale = 98.6 °F on the Fahrenheit scale.

3 **Fill in the blanks: (i) The hotness or coldness of a system is determined by its ____. (ii) The temperature of ice-cold water cannot be measured by a ____ thermometer. (iii) The unit of temperature is degree ____.**

✔ ANSWER

(i) **temperature**

(ii) **clinical**

(iii) **Celsius**

4 **The range of a laboratory thermometer is usually ____. (i) 10 °C to 100 °C (ii) −10 °C to 110 °C (iii) 32 °C to 45 °C (iv) 35 °C to 42 °C**

✔ ANSWER

(ii) −10 °C to 110 °C

5 **Four students used a laboratory thermometer to measure the temperature of water (Fig. 7.6). Who followed the correct way? (i)**

Student 1 (ii) Student 2 (iii) Student 3 (iv) Student 4

💡 SAMPLE ANSWER

(iv) Student 4 (check the picture in your book)

The correct student holds the thermometer **vertically**, with the bulb dipped in water but **not touching the bottom or sides** of the beaker, and reads it while it is in the water with the eye level with the liquid top.

6 **Colour the red column on the drawings of thermometers (Fig. 7.7) as per the temperatures: 14 °C, 17 °C, 7.5 °C.**

💡 SAMPLE ANSWER

- **14 °C:** colour from the bulb up to the 4th small mark above 10 °C.
- **17 °C:** colour up to the 7th small mark above 10 °C (3 marks below 20 °C).
- **7.5 °C:** colour up to the middle between the 7 °C and 8 °C marks.

(Each small division = 1 °C; check the scale of the drawing in your book.)

7 **Observe the part of the thermometer in Fig. 7.8: (i) What type of thermometer is it? (ii) What is its reading? (iii) What is the smallest value it can measure?**

💡 SAMPLE ANSWER

(i) A **laboratory thermometer** (it has a liquid column and a Celsius scale).

(ii) Read the mark where the top of the red liquid column stops (check the picture in your book).

(iii) Smallest value = (difference between two numbered marks) ÷ (number of small divisions between them); for 10 °C with 10 divisions it is **1 °C**.

8 **A laboratory thermometer is not used to measure our body temperature. Give a reason.**

✔ ANSWER

The liquid level starts to fall as soon as it is taken out of the mouth (or armpit), so the correct reading cannot be taken. Also, it is long and breaks easily, so it is not safe to put in the mouth.

9 **Vaishnavi's body temperature for three days (Table 7.4):**

Day	7am	10am	1pm	4pm	7pm	10pm
One	38.0	37.8	38.0	38.0	40.0	39.0
Two	38.6	38.8	39.0	39.0	39.0	38.0
Three	37.6	37.4	37.2	37.0	36.8	36.6

(all in °C)

(i) What was her highest recorded temperature? (ii) On which day and at what

time? (iii) On which day did her temperature return to normal?

✔ ANSWER

(i) **40.0 °C**

(ii) **Day One at 7 pm**

(iii) **Day Three** (it came down to 37.0 °C at 4 pm and stayed normal).

10 **If you have to measure the temperature 22.5 °C, which of the three thermometers (Fig. 7.9) will you use? Explain.**

💡 SAMPLE ANSWER

Use the thermometer whose **smallest division is 0.5 °C** and whose range includes 22.5 °C (check the picture in your book).

A thermometer whose smallest division is 1 °C or more cannot show 22.5 °C correctly.

11 **The temperature shown by the thermometer in Fig. 7.10 is (i) 28.0 °C (ii) 27.5 °C (iii) 26.5 °C (iv) 25.3 °C**

💡 SAMPLE ANSWER

(ii) 27.5 °C (check the picture in your book)

12 **A laboratory thermometer has 50 divisions between 0 °C and 100 °C. What does each division of this thermometer measure?**

✔ ANSWER

Temperature difference = 100 °C − 0 °C = 100 °C

Number of divisions = 50

One division = 100 ÷ 50 = **2 °C**

13 **Draw the scale of a thermometer in which the smallest division reads 0.5 °C (only the part between 10 °C and 20 °C).**

💡 SAMPLE ANSWER

- Mark 10 °C and 20 °C as big marks, with 15 °C as a medium mark.
- From 10 to 20 there is 10 °C; 10 ÷ 0.5 = **20 small divisions**.
- So draw 2 small divisions for every 1 °C: 10, 10.5, 11, 11.5 ... 19.5, 20 °C.

14 **Komal tells you that she has a fever of 101 degrees. Does she mean it on the Celsius scale or Fahrenheit scale?**

✔ ANSWER

Fahrenheit scale. Normal body temperature is 98.6 °F, so 101 °F is a fever. 101 °C is not possible for a human body (it does not go above 42 °C).

📘 Learning further

1 **Find out how the body temperature of animals such as cat, dog, horse, camel, cow and buffalo is measured.**

💡 SAMPLE ANSWER

Vets usually measure animal body temperature with a digital thermometer placed in the rectum (back passage); infrared thermometers are also used for some animals.

Normal temperature of many animals is higher than ours (about 38–39 °C for a dog, cat or cow).

2 **Find out which places in India are usually regarded as the coldest and hottest, and their minimum and maximum recorded temperatures.**

💡 SAMPLE ANSWER

- Coldest: **Dras** (Ladakh); winter temperatures can go below −40 °C.
- Hottest: **Phalodi** (Rajasthan); about 51 °C was recorded in May 2016.

(Check the latest records from IMD.)

3 **Make a table of the planets, their distances from the Sun (increasing order) and their temperatures. Does average temperature decrease with distance from the Sun? If not, for which planet and why?**

💡 **SAMPLE ANSWER**

Planet	Average temperature
Mercury	about 167 °C
Venus	about 464 °C
Earth	about 15 °C
Mars	about –65 °C
Jupiter	about –110 °C
Saturn	about –140 °C
Uranus	about –195 °C
Neptune	about –200 °C

Mostly yes, but **Venus** is hotter than Mercury though farther from the Sun, because its thick carbon dioxide atmosphere traps heat.

4 **Hang a room thermometer (1) and keep a laboratory thermometer in a beaker of water (2) near it. Take readings 3 times a day (first period, lunch, last period) for two weeks. Compare and conclude.**

💡 **SAMPLE ANSWER**

Time	Thermometer 1 (room)	Thermometer 2 (water)
First period	24 °C	23 °C
Lunch break	28 °C	26 °C
Last period	29 °C	27.5 °C

Conclusion: Both readings rise during the day. Water temperature changes more slowly than air temperature, so it stays a little lower when the room warms up.

🎉 **Chapter 7 done! Revise the answers once more tomorrow.**